

Practical 11 Questions

16 April 2026

- Find the propensity of alpha helices using the following sequence and secondary structure assignments.

Sequence: LGASGIAAFAFGSTAILIILFNMAAEVHFDPLQFFRQFFWLGLYPPKAQY
Sec. str: XX

Sequence: GMGIPPLHDGGWWLMAGLFMTLSLGSWWIRVYSRARALGLGTHIAWNFAA
Sec. str: XX

Sequence: AIFFFVLCIGCIHPTLVGSWSEGVFPGIWPIDWLTAFSIRYGNFYCPWH
Sec. str: HHH

Sequence: GFSIGFAYGCGLLFAAHGATILAVARFGGDREIEQITDRGTAVERAALFW
Sec. str: HHH

- Find the propensity of alpha helices manually for the sequence in question 1.
- Using the rules for helices and strands, identify the helical and strand segments in the following sequence

KVFGRCELAAAMKRHGLDNRYGYS LGNWVCAAKFESNFNTQATNRNTDGSTDY GILQINSRWWCNDGRTPGSRNL CN
IPCSALLSSDITASVNC AKKIVSDGNGMNAWVAWRNRCKGTDVQAWIRGCRL

Hint: **Helix:** Assign 1 for H_α and h_α ; 0.5 for I_α ; 0 for

i_α ; -1 for B_α and b_α ; identify 6-residue segments with score more than or equal to 4; extend it until the actual value (Table 5.2) for last four residues is less than 4. Continue the search.

Strand:

Assign 1 for H_β and h_β ; 0.5 for I_β ; 0 for i_β ; -1 for B_β

and b_β ; identify 5-residue segments with score more than or equal to 3; extend it until the actual value for the last three residues is less than 3.

For conflicting situation: compare the values and assign the secondary structure based on the highest value

TABLE 5.2 Chou-Fasman parameters

Residue	P_α	Residue	P_β
Glu	H_α 1.53	H_β Met	1.67
Ala	1.45	Val	1.65
Leu	1.34	Ile	1.60
His	h_α 1.24	h_β Cys	1.30
Met	1.20	Tyr	1.29
Gln	1.17	Phe	1.28
Trp	1.14	Gln	1.23
Val	1.14	Leu	1.22
Phe	1.12	Thr	1.20
Lys	I_α 1.07	Trp	1.19
Ile	1.00	I_β Ala	0.97
Asp	i_α 0.98	i_β Arg	0.90
Thr	0.82	Gly	0.81
Ser	0.79	Asp	0.80
Arg	0.79	b_β Lys	0.74
Cys	0.77	Ser	0.72
Asn	b_α 0.73	His	0.71
Tyr	0.61	Asn	0.65
Pro	B_α 0.59	Pro	0.62
Gly	0.53	B_β Glu	0.26

- Verify one of the helical and strand segments, manually.

Deadline: 22 April 2026